

BEDROCK ESG — PROJECT BIBLE v2.4

For: Stakeholders, Partners, and Team Members
Version: 2.4 — Trinity Protocol | **Updated:** June 16, 2026
Live at: bedrockesg.com (<https://bedrockesg.com>)

WHAT IS BEDROCK ESG?

Bedrock ESG is an **institutional-grade carbon verification platform** that provides forensic-level proof of environmental impact across three industrial verticals:

- **Asphalt Recycling** — Tracking CO₂ savings from Reclaimed Asphalt Pavement (RAP)
- **Biochar Carbon Removal** — Measuring and certifying permanent carbon dioxide removal (CDR)
- **Metal Recovery** — Quantifying emissions avoided through scrap metal recycling

The platform acts as a **digital notary** — it doesn’t just calculate carbon savings, it proves them through multi-layer verification, blockchain anchoring, and forensic document analysis.

THE CORE PROMISE: “ZERO GREED”

Every verified asset settlement follows the **70/20/10 Universal Split**:

Share	Recipient	Description
70%	Asset Sovereign	The verified asset holder — the person or entity who owns the carbon credit or physical asset
20%	Platform Processor	Bedrock ESG’s operating share for verification, hosting, and compliance
10%	Public Resilience	Community allocation — funds directed to public infrastructure and sustainability

This split is **mathematically enforced** in the codebase. It cannot be changed without triggering a system-level violation error.

HOW VERIFICATION WORKS

The Sentinel Engine — Three-Pillar Scoring

Every document, file, or sensor reading submitted to Bedrock ESG passes through the **Sentinel Engine**, which scores it across three independent pillars:

Pillar	What It Checks	Minimum Score
Document Integrity	Are files authentic? Do their internal signatures match their declared format?	95%
AI Confidence	Can our AI parser extract meaningful, consistent data from the documents?	75%
Spatial Correlation	Does the GPS location, timestamp, and metadata make geographic sense?	75%

All three pillars must pass their minimum threshold for **automatic approval**. If any pillar falls short, the submission is held for human review.

The Haversine Boundary Check

When a field operator submits data from a known facility, the system calculates the **exact distance** between the operator's GPS coordinates and the facility's registered location using the Haversine formula (the mathematical method for calculating distance on a sphere).

If the operator is more than **50 kilometers** from the facility, the submission is **immediately rejected** with a clear message: "GPS Accuracy Degraded — Please Re-Enter the Active Capture Zone."

This prevents remote spoofing of location data.

Temporal Spoofing Detection

The system compares the timestamp on the operator's device with the server's clock. If the difference exceeds **15 minutes**, the submission is flagged as potentially spoofed. This catches attempts to backdate or forward-date submissions.

THE THREE SUBMISSION TRACKS

Track A: Materials (Asphalt, Concrete, Metal)

- Upload material test reports, delivery tickets, or recycling manifests
- System extracts tonnage, material type, and ICE v4.1 emission coefficients
- Calculates CO₂ savings based on Eurobitume lifecycle analysis (A1-A3 stages)

Track B: 179D Energy (Real Estate)

- Upload IRS Form 7205, building envelope data, and energy audits

- Supports commercial and residential real estate carbon compliance
- Generates tax deduction documentation

Track C: Biochar CDR (Carbon Dioxide Removal)

- Upload pyrolysis data, feedstock documentation, and stability metrics
- Validates CDR credits against Hedera Guardian Framework
- Links to dual-blockchain attestation (Hedera + Polygon)

WHAT THE PLATFORM LOOKS LIKE

For Asset Owners (Clients)

Page	What It Does
Dashboard	Real-time cockpit ticker showing all verified assets with live value updates
Submission Portal	Drag-and-drop upload zone for documents, photos, and 3D spatial scans
Calculator	Instant CO ₂ savings estimation for asphalt recycling projects
Impact Dashboard	Visual charts showing cumulative environmental impact
Credits	Track green credits through their lifecycle: Pending → Issued → Redeemed
Onboarding	3-step wizard: Identity → Property → Telemetry setup

For Verification Agents

Page	What It Does
Agent Portal	Overview dashboard with farmer registration and pending task counts
Verification Queue	One-tap approve/reject for biochar batches and recycling logs

For Administrators

Page	What It Does
Admin Panel	Master control: approve/reject/re-scan discovery assets, execute settlements, network purge console
Discovery Feed	Intelligence feed for unlisted macro-assets (exchanges, skyscrapers, casinos)
Global Ledger	Consolidated view of all settlements and carbon impact across the platform
Audit Logs	Tamper-proof activity log with SHA-256 integrity hashes

SECURITY LAYERS

Authentication

- **Email/Password login** with encrypted password storage
- **Google Single Sign-On** for seamless institutional access
- **JWT tokens** stored as secure, httpOnly cookies (not accessible to JavaScript)

Role-Based Access Control (4 Tiers)

Role	Access Level
Founder	Full system override — all settings, all data, all actions
Admin	Full platform management — user management, settlements, discovery
Sovereign Agent	Field verification, submission approval, telemetry access
Client Owner	Submit assets, view own submissions only

Protocol Security (for API/Field App Communication)

- **Timing-safe key validation** — prevents brute-force timing attacks
- **300-second timestamp window** — requests outside this window are rejected
- **Nonce deduplication** — every request must have a unique identifier; replays are blocked

BLOCKCHAIN ANCHORING

Bedrock ESG anchors verification proofs to **three independent blockchains**:

Chain	Purpose	Status
Solana (Devnet)	Notary memo anchoring — immutable timestamp proof of every settlement	Active
Hedera (HCS)	Hashgraph Consensus Service — tamper-proof submission records	Active
Polygon (ERC-1155)	Token minting — tradeable carbon credit tokens	Active

EXITZ Token Ecosystem (Solana Mainnet)

Token	Purpose
EXITZ (base)	Core platform token (supply: 2)
EXITZ-C (Classic Vehicles)	Mobile asset verification (supply: 1)
EXITZ-A (Fine Art)	Art asset verification (supply: 1)

THE DISCOVERY PIPELINE

Bedrock ESG maintains an intelligence feed of **macro-assets** — large-scale properties and institutions that represent significant verification opportunities:

- **5 Global Exchanges:** NYSE, NASDAQ, LSE, CME, Eurex
- **4 Iconic Skyscrapers:** Burj Khalifa, Taipei 101, One World Trade, Shanghai Tower
- **3 Major Casinos:** MGM Grand, The Venetian Macao, Marina Bay Sands

Each asset carries an estimated value and a calculated **70% sovereign share**. Assets start in Pending Sovereign Review and must be explicitly approved by a Founder or Admin before settlement.

THE PURGE GATE

The Admin Panel includes a **Network Purge Console** — a destructive cleanup tool protected by multiple safety layers:

1. **Role gate** — Only Founders and Admins can access
2. **Preview scan** — Shows exactly what would be deleted before any action
3. **Typed confirmation** — The operator must type the exact word “PURGE” into a text field

4. **Modal confirmation** — A final warning dialog before execution

This ensures no accidental data destruction.

KNOWN FACILITY ZONES

The platform has four registered physical facility zones used for GPS verification:

Zone	Location	Radius
Sovereign Primary	Las Vegas, Nevada	50 km
Property Industrial	Houston, Texas	75 km
Logistics Hub	Ontario, California	40 km
Metals Recovery	Pittsburgh, Pennsylvania	60 km

Field operators submitting data must be physically within these zones for their submissions to be accepted.

EMISSION SCIENCE

Carbon calculations are grounded in two internationally recognized databases:

- **ICE Database v4.1** (University of Bath) — 25 material emission factors
- **Ecoinvent v3.10.1** (Swiss Centre for Life Cycle Inventories) — 15 process factors

A **60% biogenic variance correction** is applied for US and Brazilian markets to account for regional differences in energy grids and material sourcing.

THE HUM LISTENER

The platform monitors real-time energy market activity through the **HUM Dynamic Listener**:

- **Baseline:** 12.6 MW monitoring threshold
- **Sources:** Major exchange trading windows (NYSE, NASDAQ, LSE, CME, Eurex)
- **Trigger:** When energy market shifts exceed 5%, the system automatically anchors a verification memo to the Solana blockchain

This creates an independent, market-driven verification cadence tied to real economic activity.

PROJECT TIMELINE

Version	Date	Milestone
v1.0	—	Scrapnet 2.0 — Asphalt RAP calculator
v2.0	—	Bedrock ESG rebrand — Biochar + Metal + Vault + Blockchain
v2.2	May 2026	Settlement label rebrand (Zero Greed), Precious Metals VAL formula
v2.3	May 2026	Macro-asset discovery pipeline, cockpit ticker (superseded)
v2.4	June 2026	Trinity Protocol — RBAC, Google SSO, Audit Logs, Aethexer v1 protocol, Sentinel Engine, Forensic Ingest, Telemetry Ingest, Admin Panel, SPEXTER sensor extension, Haversine boundary enforcement

KEY CONTACTS

Role	Identity
Founder / Architect	Kato (Zer0)
System Identity	Aethexer Notary Node
Admin Email	idecade8@gmail.com
Platform	bedrockesg.com (https://bedrockesg.com)

Bedrock ESG — Institutional Carbon Verification for the Circular Economy

“Seed to Sale Chain of Custody” — Every gram verified, every dollar split fairly.